

TLE49012 and TLI49012 TLE987x Integration Example

1. Introduction

This code example provides a starting point for interfacing the TLx49012 angle sensor with a microcontroller from the MOTIX™ TLE987x family, using **SPI**.

The development boards used for this example code are:

- **TLE49012 Satellite Board;**
 - [Sensor Infineon website](#)
 - [TLE49012 Evaluation Kit Infineon website](#)
 - [TLI49012 Evaluation Kit Infineon website](#)
- **TLE987x EvalBoard**, VQFN socket:
 - TLE9871QXA20 microcontroller;
 - TLE9872QXA40 microcontroller;
 - [Evaluation Board Infineon website](#)
- **TLE9879 EvalKit V1.4:**
 - TLE9879QXA40 microcontroller;
 - [Evaluation Board Infineon website](#)

 **Important** — all example codes are made for the SPI interface

SPI is the main interface for sensor configuration such as LUT programming, interface settings or measurement settings. Depending on the sensor variant, different types of interfaces are available and specific interfaces are set by default. SPI is for every variant active right at start-up.

If the sensor requires a non-default configuration setting, the SPI interface must be kept active by sending a SPI register unlock command after $t_{\text{SPI_active}}$ and before $t_{\text{start-up}}$. Once $t_{\text{start-up}}$ has elapsed the pre-configured interface is active.

The interface type can be selected by programming `if_mode` in the `<usr_config_1_reg>` register. Please follow the sensor configuration flow as depicted in the [TLx49012 User Manual](#).

1.1. Short description

This example code performs continuous readouts of the TLx49012 angle sensor registers via SPI, at a frequency of ~20Hz.

SPI In-Frame addressing scheme is used and exemplified.

For easier interpretation and visualization, the following data is transmitted to the serial port:

- Register angle value, in LSB;
- Calculated angle value, in degrees;

Peripheral configuration is detailed in **Section 3**.

Note: The provided example is not a qualified solution and is provided "as-is".

2. Getting started

2.1. Hardware connections

The block diagram below shows the required connections between the TLx49012 angle sensor and the TLE987x evaluation boards.

Both TLE987x evaluation kits for development (TLE987x VQFN EvalBoard and TLE9879 EvalKit) have identical, labeled pin-outs.

Additional components, such as **decoupling capacitors**, are not depicted. Please refer to the [TLx49012 datasheet](#) for more details!

 ApplicationCircuit

2.2 Necessary software

In order to access all the features of the code example, the following software is required:

- **Keil  Vision5 IDE**, version 5.43.1.0.
 - IDE of choice for this example code, as it is compatible with the TLE987x microcontroller family.
 - For more details about the software, check the [official Keil MDS-ARM v5.xx website](#).
- **Infineon MCU Config Wizard V2.7.6**, for peripheral configuration.
 - This tool is free to download for users with a myInfineon account.
 - Find the software on the **Infineon Development Center (IDC) Launcher**, or [online](#).
- In case the example code flashes correctly, but the serial port does not display any information, make sure to update the J-Link driver of the on-board debugger using the **J-Link Commander**.

2.3. Project importing in Keil Vision5;

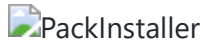
Once installed, the example code project can be imported onto your machine:

1. Create a new folder that will act as your Keil  Vision5 workspace;
2. Inside the created folder, right click, Git Bash Here, clone the code example repository;
3. Project can be opened directly, by double-clicking the `TLx49012_TLE987x_IntegrationExample.uvprojx` file, or by opening the Keil  Vision5 IDE and selecting **Project > Open Project** from the toolbar.

2.4. DFP installation in Keil Vision5

In case it is the first time using the MOTIX™ TLE987x microcontroller family, make sure to install the **Infineon TLE987x_DFP** support package:

1. Go to the toolbar and select **Project > Manage > Pack Installer**;
2. On the left side, in the **Devices** tab, search for **TLE987x** ;
3. On the right side, select the **Infineon TLE987x_DFP** package and click **Install**;



2.5. Debugger configuration

To correctly configure the debugger for your target microcontroller, follow the steps below:

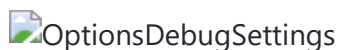
1. Go to the toolbar and select **Project > Options for Target 'src'**;
2. In the new window, select the **Device** tab and select your target microcontroller;
 - TLE9871QXA20;
 - TLE9872QXA40;
 - TLE9879QXA40;



3. Select the **Debug** tab and select from the top-right dropdown menu the **J-LINK / J-TRACE Cortex** debugger;



4. Press the **Settings** button, a new window will appear:
 - In the **J-Link / J-Trace Adapter** section, **Port** menu, select **SW** (default is JTAG);
 - In the **Connect & Reset Options** section, **Connect** menu, select **with Pre-reset** (default is Normal);



5. Press **OK** to close the window and **OK** again to close the Options window.
6. Build the project by selecting **Project > Build Target** or pressing **F7**.

2.6. Flashing the firmware

Power the TLE987x evaluation board (12V) and connect it to the PC using a micro-USB cable. In order to flash the firmware, the project must be built first, as the IDE does not automatically build before flashing.

After flashing, power cycle the board, or press the on-board RESET button.

3. Peripheral configuration

This chapter represents a rundown of the configuration done in the **MCU Config Wizard** for the peripherals used in the example code, and can be used as a reference or for configuring a new project:

- **SSC1**, for SPI communication with the sensor;
- **PORT**, for GPIO configuration of the SPI Chip Select pin;
- **UART2**, for UART communication with the PC.

To open the MCU Configuration Wizard, go to the toolbar and select **Tools > MCU Config Wizard V2.7.6**. In case the option is not available, restart the Keil  Vision5 IDE and try again.

- ▶ **SSC1**
- ▶ **PORT**
- ▶ **UART2**

4. Firmware

This chapter provides insight into the firmware structure and program flow.

4.1. Library organization

The Keil  Vision5 IDE does not allow for multi-layer folder hierarchy, so the file explorer inside the IDE does not reflect the actual folder structure of the project.

Inside the project, the folder `src` houses all the functionalities of this example code:

- `Controller` folder contains all the microcontroller-specific and peripheral functions:
 - `TLE987x_UART` folder contains the UART serial port data formatting/transmission functions;
 - `TLE987x_SPI` folder contain the SPI initialization sequence and the low-level SPI data transfer function;
 - `TLE987x_GPIO` folder contains GPIO abstraction functions and the NCS pin defines;
- `Sensor` folder contains TLx49012-specific information:
 - `TLx49012.c/h` contain definitions particular to the sensor, as well as the initialization sequence (soft-fusing, disabling CRC checks etc.);
 - `Interface` folder contains the high-level SPI in-frame data transfer functions, complete with 32-bit command generation and CRC.

4.2. Initialization

At the beginning of the program, the function 'TLE_init()' initializes the peripherals with the settings applied in **MCU Config Wizard**, particularly focusing on hardware connections.

This function is present by default upon creating a new project using the Keil Vision5 toolchain and the official DFP's.

4.3. Available functions

This chapter provides a list of the functions available in this example code.

void TLx49012_Init(void)

This function initializes the sensor by sending SPI commands.

The CRC LUT is populated with values, for faster computation when CRC calculus is needed.

A 550us delay is applied to ensure the SPI bus is active, assuming the sensor has just been powered on.

The first SPI command unlocks the internal registers, so that new data can be written.

Next, the Bitmap CRC checks are disabled by writing to the `STAT_EN_1` register.

A read-back verification is performed; if the sensor does not respond correctly, execution is halted with an assertion error.

The sensor is then soft-configured by writing to the `USR_CONFIG_1` register.

Finally, the sensor is reset from VM memory using the `STAT_EN` register, so register contents are maintained. A 900us delay is applied to wait for SPI to become active again.

A second read-back verification checks that the configuration was correctly applied after reset; if not, execution is halted.

Sensor is ready to receive further commands upon successful completion.

uint32_t TLx49012_SPI_WriteInFrame(uint8_t addr, uint16_t data)

This function represents the high-level SPI write-in-frame sequence, as described in the sensor datasheet.

A 32-bit write command is issued to the sensor, composed of the address, WRITE bit, 16-bit data (LSB-first) and calculated CRC.

Sensor response is received in the same SPI transfer frame.

`uint8_t addr` - Register address to which data is written.

`uint16_t data` - Data to be written to sensor register.

Returns `uint32_t` sensor response.

uint32_t TLx49012_SPI_ReadInFrame(uint8_t addr, bool clearStatus)

This function represents the high-level SPI read-in-frame sequence, as described in the sensor datasheet.

A 32-bit read command is issued to the sensor, composed of the address, READ bit, 0x00 byte, device status clear byte, and calculated CRC.

If the device status byte is all 1's (0xFF), the device status is cleared. Sensor response is received in the same SPI transfer frame.

`uint8_t addr` - Register address from which data is read.

`bool clearStatus` - Signals whether device status is cleared or not upon command completion.

Returns `uint32_t` sensor response.

`uint16_t TLx49012_GetAngleLSB(void)`

This function reads the content of the register at address 15.

From the 32-bit sensor response, the status and CRC bytes are ignored.

Returns `uint16_t` angle value in LSB.

`void TLE987x_UART_SendAngleInfo(uint16_t angle)`

This function displays on the serial port the register angle value [LSB] and the calculated angle value [degrees].

`uint16_t angle` - angle value in LSB to be converted to degrees, both values sent to serial port.

5. Implementation Example

This section provides the code for ~20Hz continuous readout of the TLx49012 angle sensor on the MOTIX™ microcontroller family.

Angle information (LSB and degrees) can be visualized using any serial port monitor, like hterm or Tera Term:

- **Baudrate:** 115200
- **Data bits:** 8
- **Stop bits:** 1
- **Parity:** none
- **Newline at:** CR

```

/*****
* Header Files
*****/
#include "tle_device.h"
#include "eval_board.h"
#include <stdio.h>

#include "Controller/TLE987x_UART/TLE987x_UART.h"
#include "Sensor/TLx49012.h"

/*****
* Global variables
*****/
uint16_t angle_LSB;

/*****
* Function Name: main
*****/
* Summary:
* This function provides continuous readout of the TLx49012 angle sensor.
* Peripherals are configured using the DFP-proprietary function, TLE_Init().
*
* Parameters:
* void
*
* Return:
* int
*
*****/
int main(void)
{
    // Initialize TLE9871 peripherals
    TLE_Init();

    // Initialize TLx49012 sensor
    TLx49012_Init();

    while(1)
    {
        // Service watchdog
        WDT1_Service();

        // Read 16-bit angle value
        angle_LSB = TLx49012_GetAngleLSB();

        // Send information to terminal
        TLE987x_UART_SendAngleInfo(angle_LSB);

        // ~5.5 ms for UART transmission + 44.5 ms delay = 50 ms readout period (20Hz)
        Delay_us(44500);
    }
}

```

Console output example:

[illegible]

6. Related Documents

- [TLx49012 User Manual](#) - Sensor user manual
- [XENSIV™ angle sensors](#) - Infineon angle sensors product page
- [Additional TLE49012 / TLI49012 example codes](#) - More code examples for these sensors